



Threat Intelligence Report

Assessment of Key Cyber Threats Relevant to Fadurel Technologies

Threats analysed

- Amadey (S1025)
- UNC2452

Prepared for	Fadurel Technologies
Role	Cyber Threat Intelligence Analyst
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Assessment scope	United States threat landscape

This report documents the threat intelligence investigation represented in the accompanying OpenCTI project. Findings reflect the intelligence available in the investigation dataset and the evidence captured in the project screenshots.

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Handling note

The threat ordering in this report is the ordering returned by the OpenCTI country view during the assessment. It is treated as an intelligence snapshot, not a definitive ranking of all threats affecting the United States.



1. Executive Summary

This report documents the CTI work carried out for Fadurel Technologies. The assessment looked at the threat landscape associated with the United States, where the company is headquartered.

OpenCTI was the main platform used during the work. AlienVault OTX provided additional intelligence through the connector, while MITRE ATT&CK; was used to map the observed behaviour. In the United States view, Amadey (S1025) and UNC2452 appeared first and were taken forward for deeper analysis.

Amadey was examined as a malware entity. I reviewed its associated indicators, reports, malware relationships and the Disposable Domains, Durable Hosting report. That report describes fake CAPTCHA pages using ClickFix, disposable or malicious domains, different staging methods and infrastructure associated with AS202412. The activity could progress from user execution to malware delivery, remote access and persistence.

UNC2452 was examined as a threat actor. The main campaign reviewed was CaptiveCrunch. The report describes DNS and HTTP manipulation on captive portal networks, redirection to attacker-controlled infrastructure, Microsoft 365 credential theft, device-code phishing using the Microsoft Entra ID authentication flow, and ClickFix-based malware delivery. OpenCTI also showed CornFlake and ChocoShell as associated malware and displayed eight indicators.

Key intelligence findings

The two cases show different exposure paths for a technology organisation. Amadey centres on endpoint compromise through user interaction and malware activity, while the UNC2452 case places more emphasis on identity, authentication and network-mediated credential theft.

Assessment limitation: OpenCTI relationships show how intelligence objects are connected in the knowledge base. They do not by themselves prove that every ATT&CK; technique associated with a threat was used in every campaign. Campaign-specific conclusions are therefore limited to the evidence documented in the project.

2. Investigation Requirement and Objectives

The primary intelligence requirement was to identify the top two threats targeting the country in which Fadurel Technologies is headquartered and describe each threat, including its behaviour, associated indicators and relevant campaigns or groups.

Investigation objectives

- Establish the organisational and geographic context of Fadurel Technologies.
- Assess the United States threat landscape in OpenCTI.
- Identify the leading threats represented in the available OpenCTI intelligence.
- Select the two threats for detailed investigation.
- Review behaviour, indicators, malware, campaigns, groups and infrastructure relationships.
- Map relevant activity to MITRE ATT&CK; attack patterns.
- Assess the potential relevance of the intelligence to Fadurel Technologies.
- Document supporting evidence from the OpenCTI investigation.

Assessment scope

- Threat behaviour and capabilities
- Indicators and observables
- Malware and infrastructure
- Campaigns and threat actors/groups
- Victim and relationship context where available
- MITRE ATT&CK; attack patterns
- Organisational relevance and defensive considerations

The work followed the same structure used in the investigation repository: organisational context, the national threat landscape, threat selection, detailed threat analysis, relationship analysis, organisational relevance and final reporting.

3. Fadurel Technologies Profile

For this assessment, Fadurel Technologies is treated as a technology-sector organisation headquartered in the United States of America. The project notes describe online retail, cloud computing, advertising, streaming and other technology services.

Business area	Description
Online retail	Online retail business.
Third-party sellers	Third-party sellers selling through Amazon.
Cloud computing	Cloud computing services.
Advertising	Advertising services.
Streaming	A streaming platform.
Artificial intelligence	AI services are included in the organisation description.
Logistics	Logistics services are included in the organisation description.

Why the business context matters

For the scenarios in this report, the main exposure areas are employee endpoints, identities, cloud services, online platforms and customer information. These areas could be affected by social engineering, credential theft or malware activity.



4. Collection and Methodology

OpenCTI was the central platform used for the investigation. AlienVault OTX supplied additional intelligence and observables through the connector, while MITRE ATT&CK; provided the framework for examining attack patterns and techniques.

Tool / source	Use in the investigation
OpenCTI	Primary threat intelligence platform used to organise and correlate threat actors, malware, indicators, campaigns, relationships and ATT&CK; objects.
AlienVault OTX	External threat intelligence and observable enrichment through the OpenCTI connector.
MITRE ATT&CK;	Mapping and interpretation of adversary behaviour and attack patterns.
OpenCTI screenshots	Evidence of entities, relationships, indicators, reports and platform state observed during the investigation.

Analysis workflow

1. Review the United States threat landscape in OpenCTI.
2. Select Amadey and UNC2452 from the first two displayed results.
3. Review associated campaigns, indicators, malware and infrastructure.
4. Map broader threat relationships to ATT&CK;.
5. Separate campaign-specific ATT&CK; evidence from broader knowledge-graph relationships.
6. Assess relevance to Fadurel Technologies.

Evidence handling

Screenshots captured during the work are treated as the primary evidence. Where OpenCTI truncated an indicator value or label, the report keeps the truncated value rather than filling in the missing text.



5. OpenCTI Deployment and Collection Environment

The CTI investigation was conducted in a self-hosted OpenCTI environment running on Ubuntu Server inside VMware. The deployment notes document the practical work required to install OpenCTI, configure its supporting services and control the intelligence ingestion workload.

Initial and final virtual machine configuration

Parameter	Initial	Final
Operating system	Ubuntu Server 24.04.5 LTS Live Server AMD64	Ubuntu Server 24.04.5 LTS Live Server AMD64
Virtualisation	VMware	VMware
RAM	6 GB	8 GB
Disk	60 GB	100 GB
Network	NAT / shared host address	NAT / shared host address

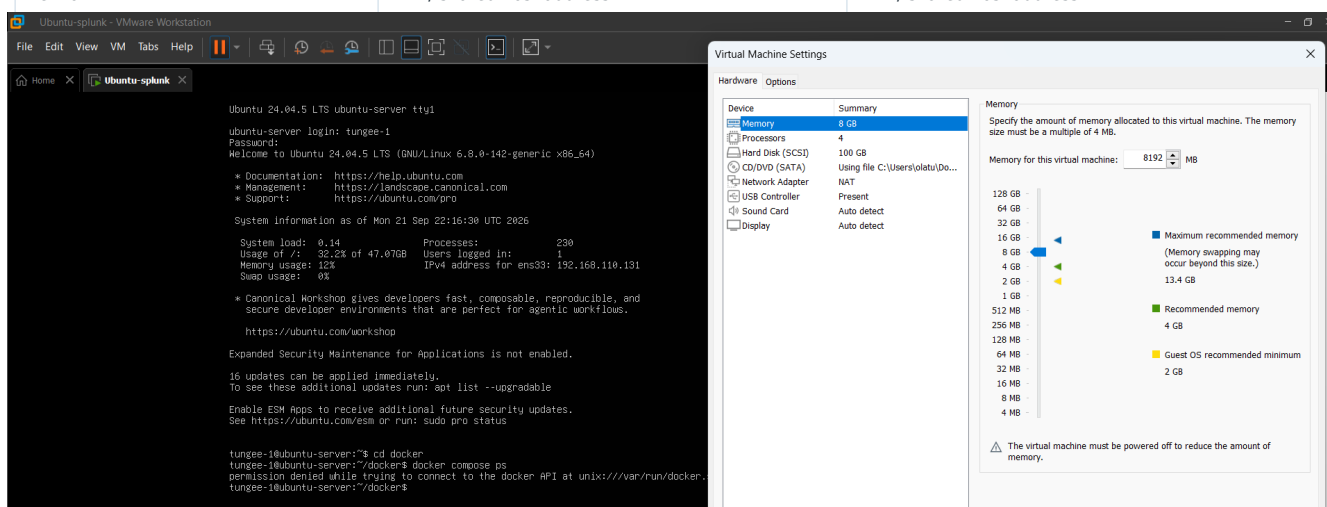


Figure 1. Final Ubuntu VM configuration used for the OpenCTI environment.

The installation required troubleshooting because the Docker Compose services took a long time to download and the VM could not comfortably handle the workload at once. Services were brought up progressively and their health was checked from the Ubuntu CLI.

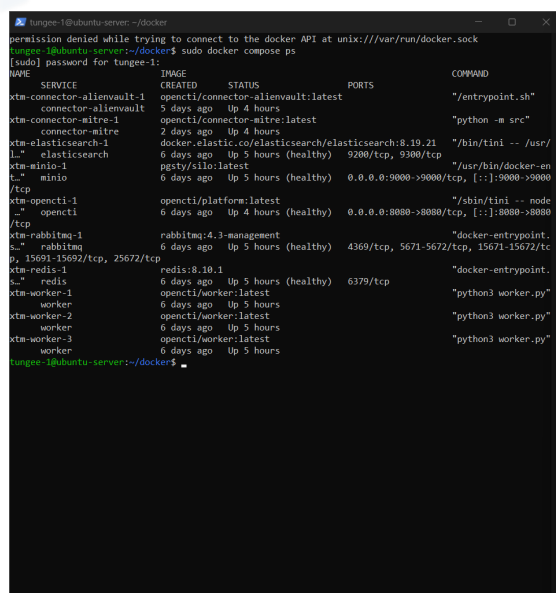


Figure 2. Docker Compose services and OpenCTI supporting components observed during the deployment.

AlienVault OTX ingestion

The AlienVault OTX connector was enabled to bring external intelligence into OpenCTI. The Integrations page showed the processing queue and ingestion rate. During collection, the queue grew while OTX continued to add data.

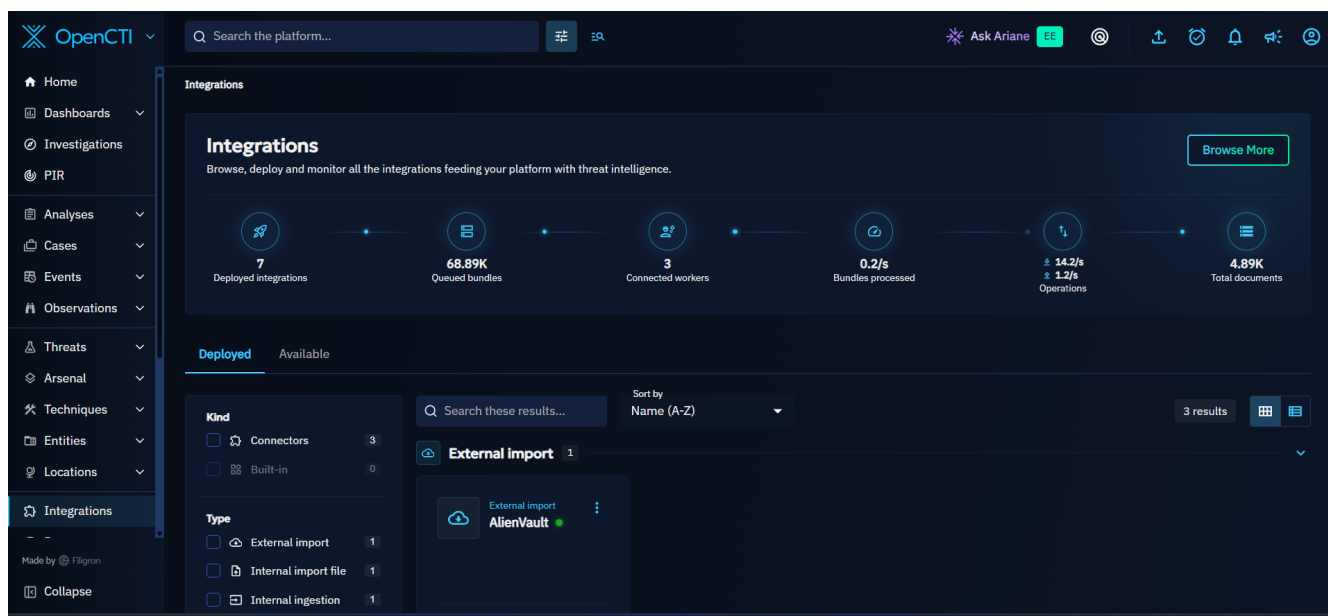


Figure 3. AlienVault OTX integration and ingestion queue observed in OpenCTI.

The first OTX ingestion covered January 2026 through September 2026. That volume put too much pressure on the home lab, affected OpenCTI performance and contributed to RabbitMQ becoming unhealthy. I reduced the collection window to three months, increased the VM to 8 GB RAM and 100 GB storage, and stopped OTX temporarily so the backlog could clear. The refreshed three-month ingestion took about three days.

Collection lesson

One practical lesson from the lab was that CTI collection volume can become a platform constraint. Limiting the connector scope and ingestion window was necessary to keep the environment usable.

6. United States Threat Landscape

Since Fadurel Technologies is headquartered in the United States, the United States was selected in OpenCTI for the national-level view.

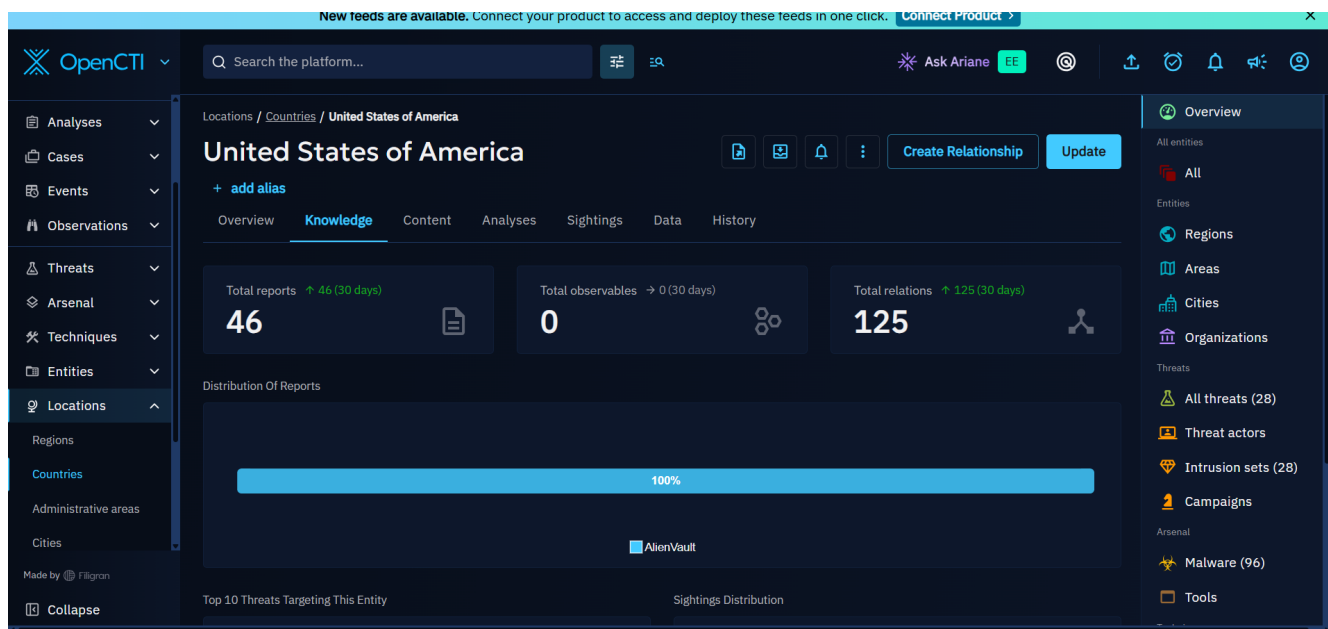


Figure 4. United States of America selected as the country of focus in OpenCTI.

The OpenCTI country view displayed 46 reports, 125 relationships and 0 observables within the displayed 30-day knowledge view at the time of the investigation.

Threats displayed in the country view

Rank	Threat
1	Amadey - S1025
2	UNC2452
3	ZigCryptoStealer
4	Cobalt Strike Beacon
5	SocGholish
6	OtterCookie
7	Red Heron
8	Psychedelic
9	UnknownKiller
10	Storm-2755

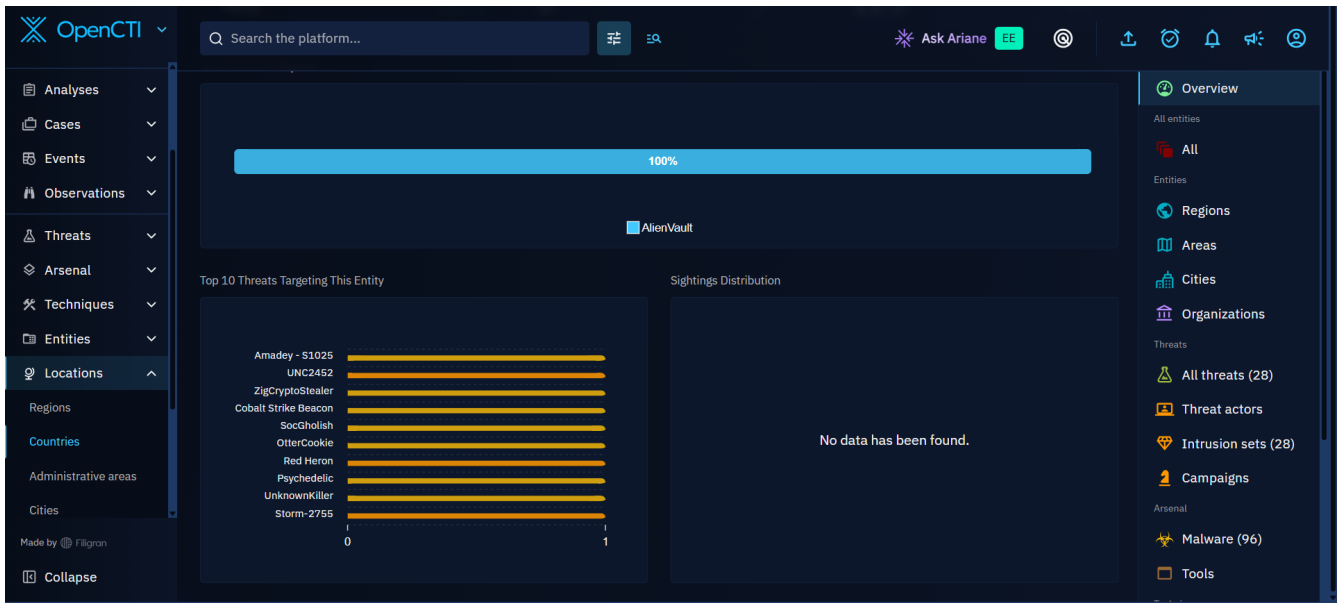


Figure 5. Top threats returned for the United States in the OpenCTI investigation.

Amadey (S1025) and UNC2452 were selected for deeper analysis based on their position in the displayed results and the availability of associated intelligence that could be investigated further.

Intelligence snapshot

The displayed order is specific to the OpenCTI dataset and view used during the assessment. It should not be interpreted as a definitive ranking of all cyber threats affecting the United States.



7. Amadey (S1025) Analysis

Amadey (S1025) appeared first in the United States threat view displayed by OpenCTI. In the knowledge base it is represented as a malware entity with associated indicators, reports, malware relationships and ATT&CK; relationships.

Type	Name	Labels	Author	Platform crea...	Analyses	Marking
Malware	Wacatac	-	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	verico-de-id.beer	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	trunnsns.beer	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	securecab.fit	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	sdntds.shop	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	rsvpopenh.one	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	pilotkadomen.club	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	pcapps.my	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR
Indicator	nttdss.shop	amadey +12	AlienVault	Sep 25, 2026	1	TLP:CLEAR

Figure 6. Amadey-related intelligence displayed in OpenCTI.

7.1 Associated reports

The Amadey page showed four related analyses/reports: Disposable Domains, Durable Hosting; VectraRAT: An Undocumented Full-Stack MaaS; Operation STANDOFF: A Campaign Hiding C2; and StealC and Amadey: Breaking Down Infostealers. I selected Disposable Domains, Durable Hosting for deeper review because it provided the clearest infrastructure and attack-chain context.

Type	Title	Author	Creators	Labels	Original cr...	Status	Marking
Report	Disposable Domains, Durable Hosting	AlienVault	admin	amadey +12	Sep 23, 2026	new	TLP:CLEAR
Report	VectraRAT: An Undocumented Full-Stack MaaS	AlienVault	admin	amadey +7	Sep 16, 2026	new	TLP:CLEAR
Report	Operation STANDOFF: A Campaign Hiding C2	AlienVault	admin	active-dire... +14	Jul 21, 2026	new	TLP:CLEAR
Report	StealC and Amadey: Breaking down infostealers	AlienVault	admin	access brok... +11	Jun 24, 2026	new	TLP:CLEAR

Figure 7. Related reports and analyses shown for Amadey in OpenCTI.

7.2 Indicators

The Amadey review included domains, IP addresses and other indicators associated with the activity shown in OpenCTI. The following values were recorded in the project notes:

verico-de-id.beer
trunnsns.beer

securecab.fit
sdntds.shop
rsvpopenh.one
pilotkadomen.club
pcapps.my
nttdss.shop
kerosand.net
idverification-code.beer
id-verif-code.info
approvalrequest-api.com
gettrack.my
fraudtechnology.com
193.202.84.17
176.65.144.127

The screenshot shows the OpenCTI interface with a list of indicators. The left sidebar contains navigation options: Threats, Arsenal, Techniques, Entities, Locations, Regions, Countries, Administrative areas, Cities, Positions, Integrations, Data, and Settings. The main panel displays a table of indicators, all associated with the 'amadey' group and a score of '+12'. The indicators are listed with their names, types, and associated entities.

Indicator	Type	Score	Entity	Date	Count	Action
dnsnewtds.shop	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
DarkGate	Malware	-	AlienVault	Sep 25, 2026	1	TLP: CLEAR
d77bc0bb3018b6cc834c1af1eefaa1c0b906314	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
codeverificatorcl.info	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
clnsdns.beer	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
clacndjsvulnarbi.beer	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
chekbrow.beer	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
cdn.claritydelivr.com	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
cdn-plugin-js.beer	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
cdn-2faclov.sbs	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
catholicsma.com	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR

Figure 8. Amadey indicator list in OpenCTI.

The screenshot shows the OpenCTI interface with a list of indicators. The left sidebar contains navigation options: Threats, Arsenal, Techniques, Entities, Locations, Regions, Countries, Administrative areas, Cities, Positions, Integrations, Data, and Settings. The main panel displays a table of indicators, all associated with the 'amadey' group and a score of '+12'. The indicators are listed with their names, types, and associated entities.

Indicator	Type	Score	Entity	Date	Count	Action
Matanbuchus	Malware	-	AlienVault	Sep 25, 2026	1	TLP: CLEAR
marketing080company.one	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
lockpopclickgetfile.monster	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
lcates-vs.beer	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
kerosand.net	Indicator	+20	AlienVault	Sep 25, 2026	2	TLP: CLEAR
idverification-code.beer	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
id-verif-code.info	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
http://approvalrequest-api.com	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
gettrack.my	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
fraudtechnology.com	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
framework-css-styles-js.beer	Indicator	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR

Figure 9. Detailed information for an Amadey-associated indicator.

7.3 Disposable Domains, Durable Hosting

The selected report describes several malicious activity chains observed over about five months. The activity used infrastructure associated with AS202412 / OMEGATECH LTD, registered in Seychelles. The chains used different delivery and staging methods, including cloud storage, disposable domains, compromised websites, trojanized installers and blockchain-resolved C2 infrastructure.

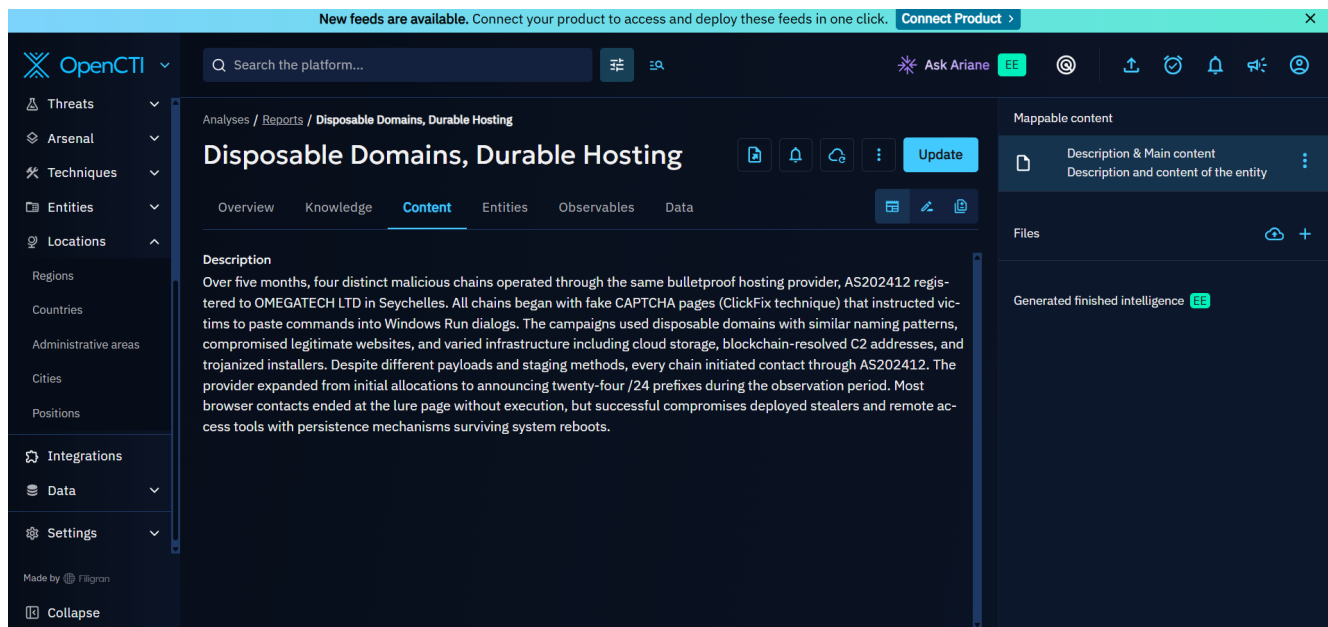


Figure 10. Disposable Domains, Durable Hosting report examined during the Amadey investigation.

The evidence reviewed here links the activity to infrastructure associated with AS202412 during the observed period. This does not mean that every malicious domain was hosted on one physical server. The relationship is to the shared provider/infrastructure.

ClickFix and fake CAPTCHA

Fake CAPTCHA pages were used as part of the social-engineering chain described in the report. The victim was told to copy and execute a command through the Windows Run interface, so successful compromise depended on the victim following the instructions.

Victim interaction and post-execution activity

The report notes that many visitors stopped at the initial lure page. Where a victim followed the instructions, the chain could continue to malware delivery and execution. The observed activity included stealers and remote access trojans, with persistence allowing successful infections to remain active after a restart.

Infrastructure expansion

The report also records expansion of the provider infrastructure to multiple /24 prefixes. Because disposable domains and other infrastructure can change, individual indicators may become obsolete. The relationships between the infrastructure, reports and malware are therefore useful when reviewing the activity.

7.4 Amadev attack chain

```

Victim visits malicious page
↓
Fake CAPTCHA / ClickFix lure
↓
Victim copies command
↓
Windows Run / command execution
↓
Malware delivery or execution
↓
Stealer / RAT activity
↓
Persistence
↓
Continued access or activity
  
```

7.5 Malware relationships

The OpenCTI investigation provided malware-related relationships beyond the Amadev entity itself. The project notes record Amadev, Amatera, Wacatac, Matanbuchus and DarkGate in the campaign relationship view. These are treated as knowledge-graph relationships rather than proof that every malware family participated in every observed chain.

Type	Name	Entity	Score	Source	Date	Count	Actions
Indicator	approvalrequest-api.com	amadev	+21	AlienVault	Sep 24, 2026	2	TLP: CLEAR
Malware	Amatera	-		AlienVault	Sep 24, 2026	2	TLP: CLEAR
Malware	Amadev - S1025	-		AlienVault	Sep 22, 2026	4	TLP: CLEAR
Indicator	alianszeg.shop	amadev	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
Indicator	ai-nexora.sbs	amadev	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
Indicator	a410c89db9140ed9dff55bf00b0338fbdffcc70f	amadev	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
Indicator	9a736f4812b485f9cf5b1332a791b205b5135a	amadev	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
Indicator	81ecbf004dc9dbf8ea4c50bde1ed55806fb5fd6	amadev	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
Indicator	193.202.84.17	amadev	+12	AlienVault	Sep 25, 2026	1	TLP: CLEAR
Indicator	176.65.144.127	amadev	+20	AlienVault	Sep 25, 2026	2	TLP: CLEAR

Figure 11. Malware-related intelligence associated with the Amadev investigation.

7.6 MITRE ATT&CK mapping

The OpenCTI knowledge view links Amadey to a broad range of ATT&CK; attack patterns across the adversary lifecycle. I kept this broader relationship map separate from the campaign-specific mapping.

The knowledge view covered Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration and Impact.

An OpenCTI relationship indicates that the attack pattern is associated with the Amadey knowledge object. It does not prove that every technique was used in every Amadey campaign.

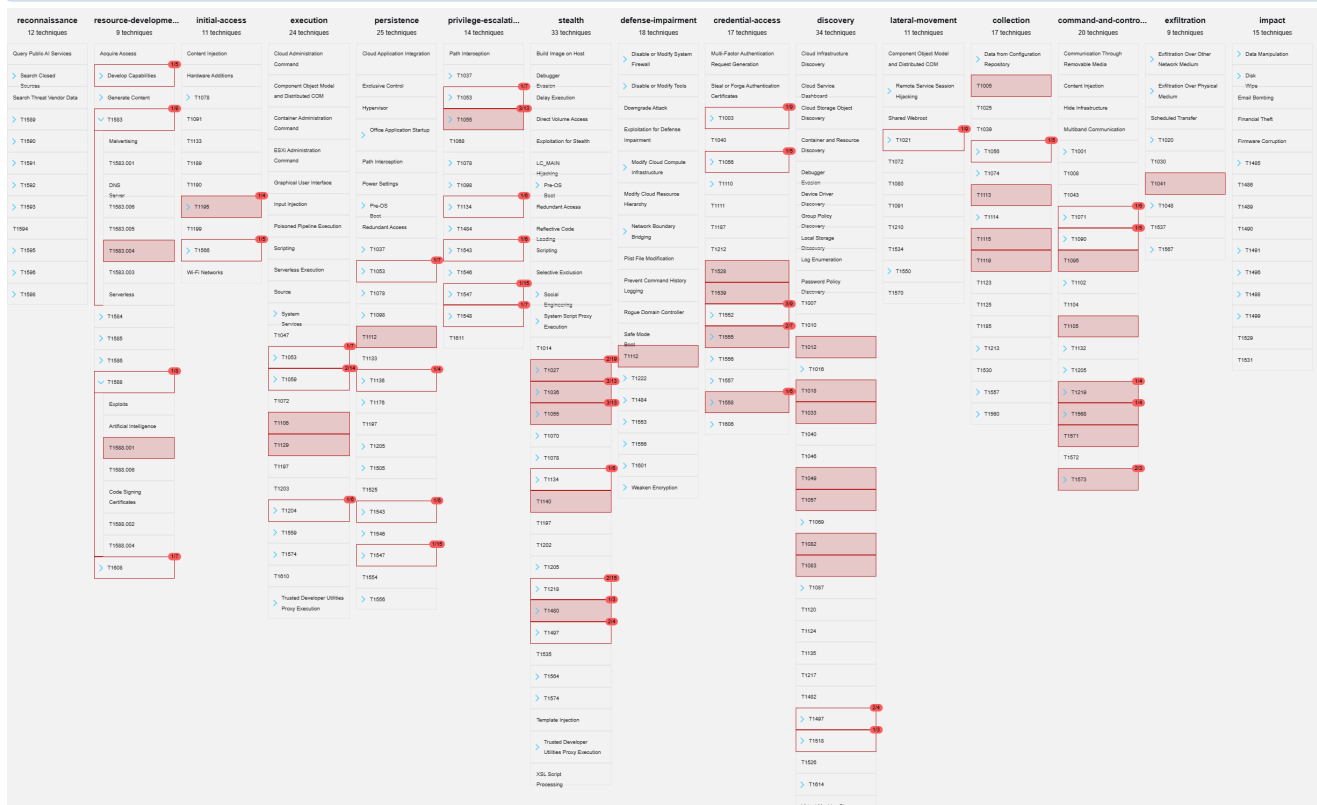


Figure 12. MITRE ATT&CK; attack patterns associated with Amadey as displayed in OpenCTI.

Campaign-specific mapping: Disposable Domains, Durable Hosting

Attack Stage	MITRE ATT&CK; ID	Attack Pattern	Relationship to Observed Campaign
Infrastructure Acquisition	T1583.001	Acquire Infrastructure: Domains	The campaign used disposable/malicious domains as part of the attacker-controlled infrastructure associated with AS202412.
Web Infrastructure	T1608.004	Stage Capabilities: Drive-by Target	The attacker prepared malicious web resources through which victims were directed into the attack chain.
Initial Access / Victim Interaction	T1204.004	User Execution: Malicious Copy and Paste	The fake CAPTCHA/ClickFix page instructed victims to copy and paste a malicious command.
Command and Control	T1568	Dynamic Resolution	Attacker infrastructure was used to support the resolution or location of infrastructure associated with C2 activity.
Remote Access	T1219	Remote Access Software	Remote-access capability was associated with the observed malware activity.
Persistence	T1547	Boot or Logon Autostart Execution	The malware demonstrated persistence that allowed it to remain active after system restart.
Exfiltration	T1041	Exfiltration Over C2 Channel	Relevant where collected information was transmitted through the C2 channel.

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Resource Development	T1583	Acquire Infrastructure	Obtaining infrastructure that can support malicious operations.
Resource Development	T1583.004	Acquire Infrastructure: Server	Obtaining or renting servers for activities such as hosting, staging or C2.
Resource Development	T1588	Obtain Capabilities	Obtaining capabilities needed for an operation rather than developing everything internally.
Resource Development	T1588.001	Obtain Capabilities: Malware	Obtaining malware as a capability that can be used during an operation.
Resource Development	T1608	Stage Capabilities	Preparing or placing capabilities on infrastructure so they can later be used against targets.
Initial Access	T1185	Browser Session Hijacking	Abusing a browser session to intercept or inherit authenticated browser information such as cookies or sessions.
Initial Access	T1506	Web Session Cookie	Using stolen web-session cookies to authenticate to web applications without repeating the normal authentication process.
Execution	T1053	Scheduled Task/Job	Using scheduled tasks or jobs to execute malicious code at a particular time or repeatedly.
Execution	T1059	Command and Scripting Interpreter	Using command shells or scripting interpreters to execute commands or scripts.
Execution	T1106	Native API	Using operating-system APIs to perform actions such as creating processes or gathering information.
Execution	T1129	Shared Modules	Executing code by loading shared modules such as DLLs or shared objects.
Execution	T1204	User Execution	Relying on the victim to perform an action that causes malicious activity to execute.
Persistence	T1053	Scheduled Task/Job	Using scheduled execution to maintain recurring or persistent execution.
Persistence	T1112	Modify Registry	Modifying Windows Registry settings to support persistence, execution, configuration changes or defense evasion.
Persistence	T1136	Create Account	Creating an account that can provide continued access to a system or environment.
Persistence	T1543	Create or Modify System Process	Creating or modifying services, daemons or other system processes for persistent execution.
Persistence	T1547	Boot or Logon Autostart Execution	Configuring a system so malicious code automatically executes during boot or user logon.
Privilege Escalation	T1037	Boot or Logon Initialization Scripts	Using scripts that execute during system boot or logon, potentially providing persistence or elevated execution.
Privilege Escalation	T1053	Scheduled Task/Job	Abusing scheduled execution to run code under a more privileged context.
Privilege Escalation	T1055	Process Injection	Injecting malicious code into another process, potentially hiding execution or obtaining access to another process context.
Privilege Escalation	T1134	Access Token Manipulation	Manipulating access tokens to operate under another security context or obtain additional privileges.
Privilege Escalation	T1543	Create or Modify System Process	Creating or modifying system services/processes that may execute with elevated privileges.
Privilege Escalation	T1547	Boot or Logon Autostart Execution	Abusing autostart mechanisms that may execute with higher privileges.
Stealth	T1027	Obfuscated Files or Information	Obfuscating code, strings, files or other information to make analysis and detection more difficult.
Stealth	T1036	Masquerading	Making malicious files, processes or other objects appear legitimate.
Stealth	T1055	Process Injection	Hiding malicious execution inside another process.
Stealth	T1134	Access Token Manipulation	Manipulating security tokens to change the security context under which activity runs.
Stealth	T1140	Deobfuscate/Decode Files or Information	Decoding or decrypting information that was previously obfuscated.
Stealth	T1218	System Binary Proxy Execution	Abusing trusted system binaries to execute malicious code.
Stealth	T1480	Execution Guardrails	Using environmental conditions to determine whether malicious activity should execute.
Stealth	T1497	Virtualization/Sandbox Evasion	Detecting or avoiding virtualized, sandboxed or analysis environments.

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Defense Impairment	T1112	Modify Registry	Modifying Registry settings can be used to alter security-related configuration or otherwise impair defenses.
Credential Access	T1003	OS Credential Dumping	Obtaining credentials from operating-system locations such as credential databases or memory.
Credential Access	T1056	Input Capture	Capturing information entered by users, including potentially credentials.
Credential Access	T1528	Steal Application Access Token	Stealing application access tokens that can be used to access resources as the legitimate user or application.
Credential Access	T1539	Steal Web Session Cookie	Stealing authenticated web-session cookies so they can be reused to access applications.
Credential Access	T1552	Unsecured Credentials	Searching for credentials stored in insecure locations.
Credential Access	T1555	Credentials from Password Stores	Obtaining credentials from password stores such as browser or operating-system credential repositories.
Credential Access	T1568	Dynamic Resolution	Using dynamic mechanisms to locate or resolve attacker infrastructure.
Discovery	T1007	System Service Discovery	Enumerating services running or registered on a system.
Discovery	T1010	Application Window Discovery	Identifying open application windows to understand what applications are being used or identify security tools.
Discovery	T1012	Query Registry	Querying the Windows Registry to gather information about configuration, software or the operating system.
Discovery	T1018	Remote System Discovery	Identifying other systems on the network that may be useful for further activity.
Discovery	T1023	Shortcut Modification	Modifying shortcuts so that a malicious program or command executes when the shortcut is used.
Discovery	T1049	System Network Connections Discovery	Enumerating active network connections involving the compromised system.
Discovery	T1057	Process Discovery	Enumerating running processes on a system.
Discovery	T1082	System Information Discovery	Gathering information about the operating system, computer name, hardware or system configuration.
Discovery	T1083	File and Directory Discovery	Searching files, directories and filesystem locations for useful information.
Discovery	T1497	Virtualization/Sandbox Evasion	Checking for virtualization or sandbox indicators to determine whether the environment is suitable for execution.
Discovery	T1518	Software Discovery	Identifying installed software and software versions.
Lateral Movement	T1021	Remote Services	Using remote services or protocols to access other systems.
Collection	T1005	Data from Local System	Collecting information from the compromised local system.
Collection	T1056	Input Capture	Capturing information entered by the user.
Collection	T1113	Screen Capture	Taking screenshots of the victim's screen.
Collection	T1115	Clipboard Data	Collecting information stored in the clipboard.
Collection	T1119	Automated Collection	Automatically gathering information according to predefined collection logic.
Command and Control	T1071	Application Layer Protocol	Using application-layer protocols for command and control communication.
Command and Control	T1095	Non-Application Layer Protocol	Using protocols outside normal application-layer protocols for C2 communication.
Command and Control	T1105	Ingress Tool Transfer	Downloading files, tools or additional malware onto a compromised system.
Command and Control	T1210	Exploitation of Remote Services	Exploiting remote services to gain access to additional systems.
Command and Control	T1568	Dynamic Resolution	Dynamically resolving or locating C2 infrastructure.
Command and Control	T1571	Non-Standard Port	Using unusual ports for network communication to make traffic harder to identify or filter.
Command and Control	T1573	Encrypted Channel	Encrypting C2 communications to prevent defenders from easily inspecting the traffic.
Exfiltration	T1041	Exfiltration Over C2 Channel	Sending stolen information through an existing command-and-control channel.

7.7 Amadey - Relevance to Fadurel Technologies

Fadurel Technologies operates online retail, third-party seller, cloud computing, advertising and streaming activities. The company also has a broader technology profile including AI and logistics services. These operations rely on endpoints, identities and online systems.

A ClickFix-style compromise is relevant because an employee browsing from a company endpoint could encounter a malicious page and be persuaded to run a command. If the endpoint were compromised, the activity could progress to malware delivery, remote access, identity compromise, lateral movement and possible exposure of confidential information.

Potential impact areas

- Endpoint availability and integrity if malware spreads or disrupts affected devices.
- Customer-data confidentiality if information is collected and removed from affected systems.
- Identity compromise if employee credentials or authentication material are obtained.
- Potential disruption of online or streaming services if supporting systems are affected.
- Compromise of online business infrastructure if an attacker gains further access.

Potential Fadurel attack scenario

```
Fadurel Technologies employee
↓
Internet browsing
↓
Malicious website
↓
ClickFix / fake CAPTCHA
↓
Employee runs command
↓
Endpoint compromise
↓
Malware download
↓
Further access
↓
Identity compromise / Remote access
↓
Lateral movement
↓
Potential data exposure
```

8. UNC2452 Analysis

UNC2452 appeared second in the United States results selected from OpenCTI. In this report it is treated as a threat actor associated with credential theft, phishing, account access and maintaining access.

8.1 CaptiveCrunch campaign

The main campaign reviewed for UNC2452 was CaptiveCrunch. The OpenCTI report describes DNS and HTTP manipulation on captive portal networks at hotels, conference centers and hospitality venues, with victims redirected to attacker-controlled infrastructure.

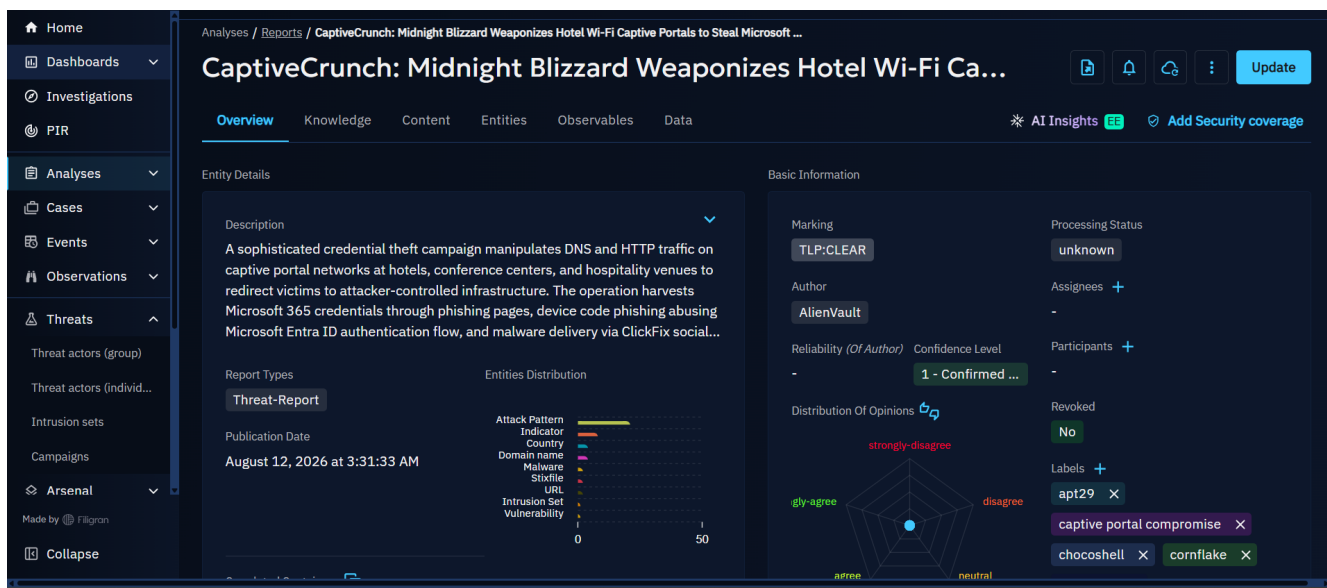


Figure 13. CaptiveCrunch report displayed in OpenCTI. The report was authored by AlienVault and shows a publication date of 12 August 2026.

The report labels include apt29, captive portal compromise, chocoshell and cornflake.

Credential theft and authentication abuse

The campaign description says that Microsoft 365 credentials are targeted through phishing pages and that the Microsoft Entra ID authentication flow is abused through device-code phishing. Identity and authentication are therefore central to this activity.

ClickFix and malware delivery

The report also identifies ClickFix social engineering as part of malware delivery. The OpenCTI investigation associates the campaign activity with CornFlake and ChocoShell.

8.2 Malware associated with UNC2452

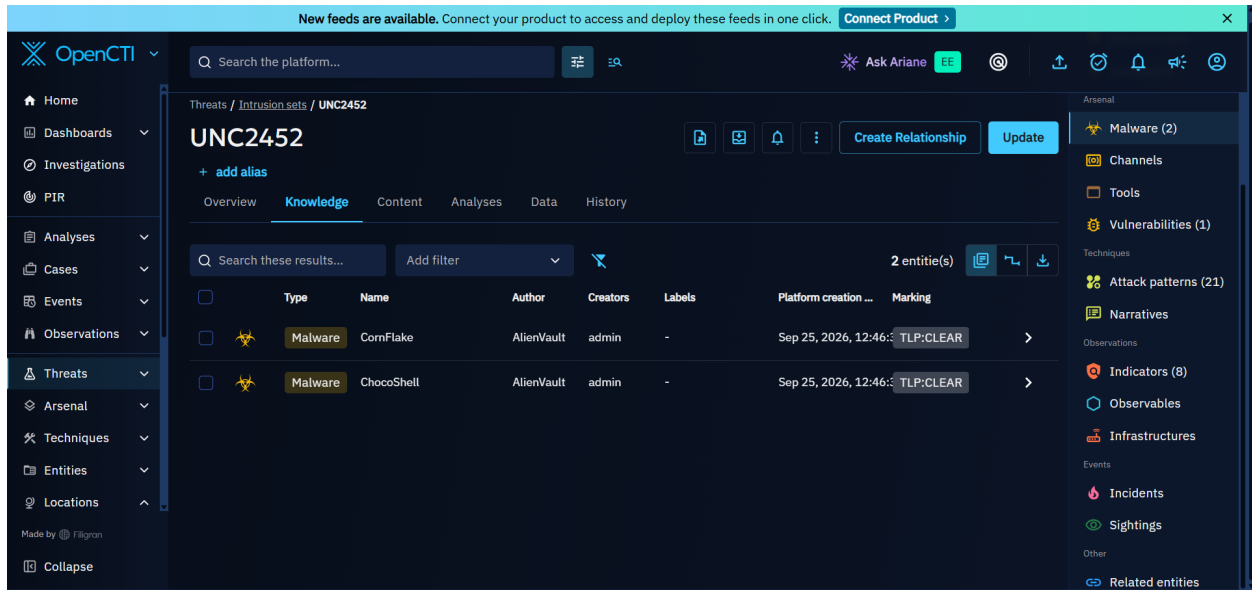


Figure 14. CornFlake and ChocoShell shown as malware entities associated with UNC2452 in OpenCTI.

Malware	Relationship in project
CornFlake	Malware associated with UNC2452 activity.
ChocoShell	Malware associated with UNC2452 activity.

8.3 Indicators

OpenCTI displayed eight indicators associated with the UNC2452 threat. Some values and labels were truncated in the interface and are retained as displayed.

Indicator	Type / Format	Label
http://213.145.86.112/t/pixel.gif	URL	apt29
http://213.145.86.112/cdn/chunks/pol...	URL	apt29
918fa52ae45ed60ba7cc8bdc99c3cbe9...	Hash	apt29
be99857449d2856dd5a84e21c8a3d5...	Hash	app p...
owa-ms365.com	Domain	app p...
ms365-live.com	Domain	app p...
ms365-device.com	Domain	app p...
m365-owa.com	Domain	app p...

OpenCTI

Search the platform...

Overview Knowledge Content Analyses Data History

Search these results... Add filter Select saved filter

Pattern	Name	Labels	Platform creati	Valid until	Marking
☐ Stix	http://213.145.86.112/t/pixel.gif	apt29 +8	Sep 25, 2026	Sep 1, 2026, 9:06:12 AM	TLP:CLEAR >
☐ Stix	http://213.145.86.112/cdn/chunks/po...	apt29 +8	Sep 25, 2026	Sep 1, 2026, 9:06:12 AM	TLP:CLEAR >
☐ Stix	918fa52ae45ed60ba7cc8bdc99c3cbe9...	apt29 +8	Sep 25, 2026	May 29, 2027, 2:41:17 PM	TLP:CLEAR >
☐ Stix	be99857449d2856dd5a84e21c8a3d5...	app p... +22	Sep 24, 2026	Jun 7, 2027, 5:18:59 AM	TLP:CLEAR >
☐ Stix	owa-ms365.com	app p... +22	Sep 24, 2026	Dec 1, 2026, 9:04:23 AM	TLP:CLEAR >
☐ Stix	ms365-live.com	app p... +27	Sep 24, 2026	Dec 1, 2026, 9:04:23 AM	TLP:CLEAR >
☐ Stix	ms365-device.com	app p... +22	Sep 24, 2026	Dec 1, 2026, 9:04:23 AM	TLP:CLEAR >
☐ Stix	m365-owa.com	app p... +22	Sep 24, 2026	Dec 1, 2026, 9:04:23 AM	TLP:CLEAR >

Home Dashboards Investigations PIR Analyses Cases Events Observations Threats Arsenal Techniques Entities Locations Collapse

Made by Filigran

Arsenal: Malware (2), Channels, Tools, Vulnerabilities (1), Techniques: Attack patterns (21), Narratives, Observations: Indicators (8), Observables, Infrastructures, Events: Incidents, Sightings, Other: Related entities

Figure 15. Eight indicators associated with UNC2452 displayed in OpenCTI.



8.4 Threat behaviour observed

- Captive portal compromise
- DNS traffic manipulation
- HTTP traffic manipulation
- Redirection to attacker-controlled infrastructure
- Microsoft 365 credential theft
- Phishing
- Device-code phishing abusing the Microsoft Entra ID authentication flow
- ClickFix social engineering
- Malware delivery

8.5 Relationship between threat, campaign, malware and indicators

UNC2452

- CaptiveCrunch campaign
 - Captive portal compromise
 - DNS / HTTP traffic manipulation
 - Microsoft 365 credential theft
 - Entra ID authentication abuse
 - ClickFix social engineering
 - 8 indicators

- Malware
 - CornFlake
 - ChocoShell



8.6 MITRE ATT&CK relationship mapping

OpenCTI associates UNC2452 with a set of ATT&CK; attack patterns across Stealth, Privilege Escalation, Collection, Execution, Credential Access and Initial Access. The relationship map below is the broader UNC2452 profile captured in the project.

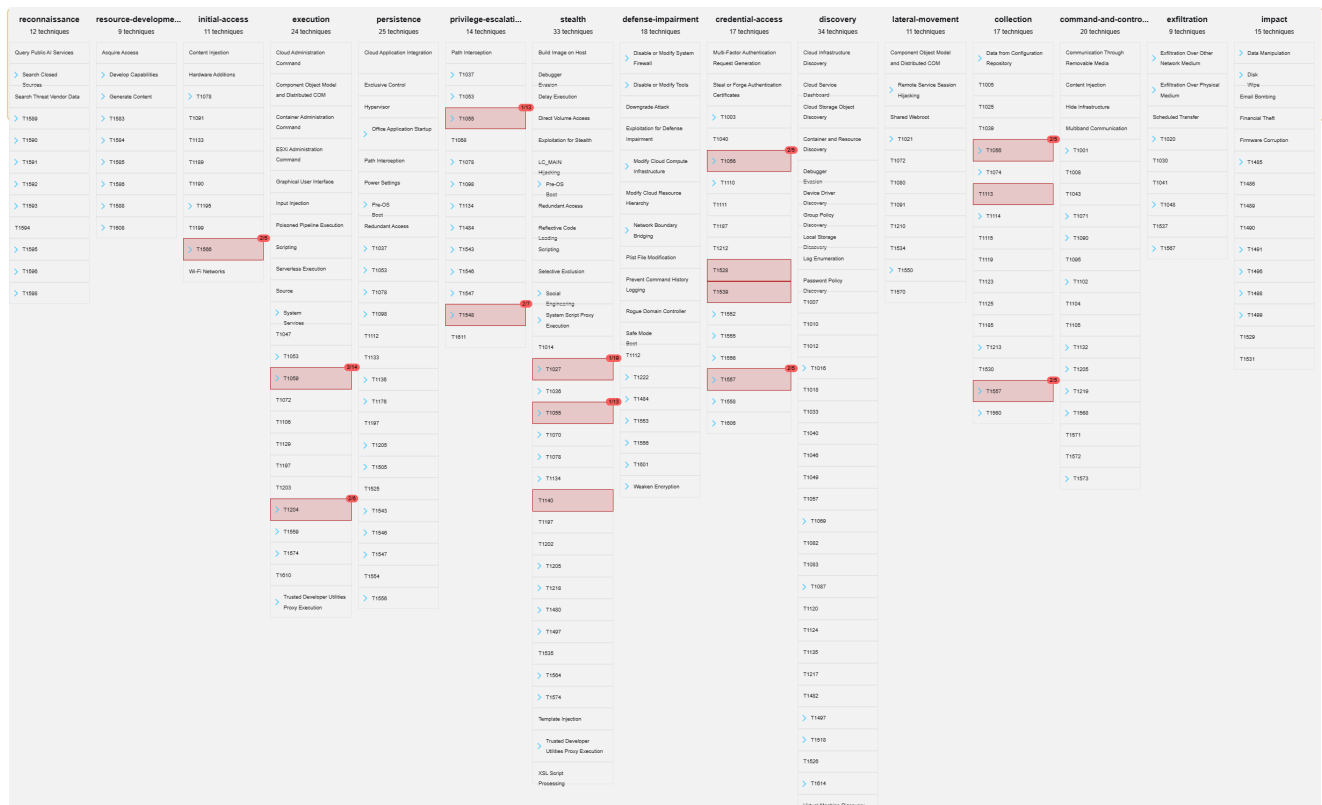


Figure 16. MITRE ATT&CK; relationships associated with UNC2452 in OpenCTI.

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Stealth	T1027	Obfuscated Files or Information	Obfuscating files, code or other information to make analysis and detection more difficult.
Privilege Escalation	T1055	Process Injection	Injecting code into another process to execute malicious code and potentially evade defenses or elevate privileges.
Collection	T1056	Input Capture	Capturing user input that may contain credentials or other information.
Collection	T1056.001	Input Capture: Keylogging	Capturing keystrokes from a victim system.
Execution	T1059	Command and Scripting Interpreter	Using command or scripting interpreters to execute commands, scripts or binaries.
Execution	T1059.001	PowerShell	Using PowerShell to execute commands or scripts.
Execution	T1059.003	Windows Command Shell	Using the Windows Command Shell to execute commands or scripts.
Collection	T1113	Screen Capture	Capturing screenshots of a victim's screen.
Stealth	T1140	Deobfuscate/Decode Files or Information	Decoding or decrypting information that was previously obfuscated or encrypted.
Execution	T1204	User Execution	Relying on the victim to perform an action that causes malicious activity to execute.
Execution	T1204.003	User Execution: Malicious File	Relying on the victim to open or execute a malicious file.
Credential Access	T1528	Steal Application Access Token	Stealing application access tokens that can be used to access resources as a legitimate user or application.
Credential Access	T1539	Steal Web Session Cookie	Stealing authenticated web-session cookies for reuse in accessing web applications.
Privilege Escalation	T1548	Abuse Elevation Control Mechanism	Circumventing mechanisms designed to control privilege elevation.
Privilege Escalation	T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control	Bypassing Windows User Account Control to execute with elevated privileges.

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Collection	T1557	Adversary-in-the-Middle	Positioning between networked devices to support activities such as credential theft, network sniffing or traffic manipulation.
Collection	T1557.002	Adversary-in-the-Middle: ARP Cache Poisoning	Manipulating ARP cache information to position the adversary between networked systems.
—	T1562	Impair Defenses	Attempting to weaken or interfere with security controls and defensive mechanisms.
—	T1562.001	Impair Defenses: Disable or Modify Tools	Disabling or modifying security tools or their configurations to reduce defensive visibility or protection.
Initial Access	T1566	Phishing	Using phishing techniques to gain access to victim systems.
Initial Access	T1566.002	Phishing: Spearphishing Link	Using malicious links to direct victims toward attacker-controlled content or resources.



Campaign-specific ATT&CK mapping: CaptiveCrunch

Attack Stage	MITRE ATT&CK; ID	Attack Pattern	Relationship to Observed Campaign
Network Positioning	T1557	Adversary-in-the-Middle	The CaptiveCrunch report describes manipulation of DNS and HTTP traffic on captive portal networks.
User Interaction	T1204	User Execution	The campaign uses social engineering to influence victim actions.
User Interaction	T1204.003	User Execution: Malicious File	To be confirmed from the underlying campaign evidence before being treated as a specific technique.
Credential Access	T1528	Steal Application Access Token	The campaign abuses the Microsoft Entra ID authentication flow; the exact token-theft mechanism should be confirmed from the underlying report.
Credential Access	T1539	Steal Web Session Cookie	To be confirmed from the underlying campaign evidence.
Initial Access	T1566	Phishing	The CaptiveCrunch report describes phishing pages used to harvest Microsoft 365 credentials.
Initial Access	T1566.002	Phishing: Spearphishing Link	To be confirmed if the underlying campaign evidence establishes delivery through a malicious link.

The campaign mapping is deliberately conservative. The CaptiveCrunch evidence directly describes DNS/HTTP manipulation, phishing pages, device-code phishing and ClickFix. Where the project notes mark a technique as requiring confirmation, it remains unconfirmed in this report.

CaptiveCrunch attack flow

```

UNC2452
↓
CaptiveCrunch campaign
↓
Captive portal network
↓
DNS / HTTP traffic manipulation
↓
Victim redirection
↓
Phishing / social engineering
■■■ Microsoft 365 credential theft
■■■ ClickFix / malware delivery
↓
Potential compromise
↓
Further access
  
```

8.7 UNC2452 - Relevance to Fadurel Technologies

The UNC2452 investigation is relevant to Fadurel Technologies because the observed CaptiveCrunch activity targets authentication and online access rather than relying only on direct malware infection. Fadurel Technologies operates cloud and online services, making identity and account security an important part of the threat picture.

If an employee or user associated with Fadurel Technologies encountered an attack similar to CaptiveCrunch, a compromised or manipulated captive portal could redirect the victim to attacker-controlled infrastructure. A phishing page or device-code phishing flow could then be used to target Microsoft 365 authentication information.

Potential attack progression

1. Employee or user connects to a captive portal network.
2. DNS or HTTP traffic is manipulated.
3. Victim is redirected to attacker-controlled infrastructure.
4. Victim is presented with phishing or social-engineering content.
5. Microsoft 365 credentials or authentication information are targeted.
6. Device-code phishing may be used to abuse the Microsoft Entra ID authentication flow.
7. ClickFix may be used for malware delivery.
8. Further access to the compromised account or endpoint may occur.
9. Additional systems or information may then be targeted.
10. Confidential information could potentially be accessed or exposed.

Potential impact areas

- Identity compromise through stolen credentials or authentication information.
- Unauthorised access to Microsoft 365 or other organisational resources.
- Exposure of confidential customer or business information.
- Unauthorised access to cloud-based resources.
- Further compromise of organisational systems.
- Potential disruption to online business services.
- Potential compromise or misuse of systems supporting online retail and streaming services.

Potential Fadurel scenario

```

Fadurel Technologies employee
↓
Connects to network
↓
Captive portal
↓
DNS / HTTP manipulation
↓
Redirected to attacker infrastructure
↓
Phishing page / authentication attack
├─ Microsoft 365 credential theft
├─ Device-code phishing
↓
Authentication compromise
  
```

↓
Account access / ClickFix
↓
Malware delivery
↓
Endpoint access
↓
Further access
↓
Sensitive data
↓
Data exposure



9. Analyst Assessment and Defensive Considerations

The investigation produced two related but distinct threat exposure patterns. Amadey activity is strongly associated in the project with endpoint compromise through user interaction, malware delivery, persistence and remote-access or information-stealing activity. UNC2452 activity, through CaptiveCrunch, places more emphasis on network-mediated redirection, credential theft and abuse of authentication flows, with ClickFix-based malware delivery also present.

Area	Amadey	UNC2452 / CaptiveCrunch
Primary intelligence object	Malware (S1025)	Threat actor
Investigated campaign	Disposable Domains, Durable Hosting	CaptiveCrunch
Initial user interaction	Fake CAPTCHA / ClickFix	Captive portal / phishing / social engineering
Credential or identity exposure	Associated with broader malware relationships and potential identity compromise	Microsoft 365 credentials and Entra ID authentication flow are directly described in campaign evidence
Malware relationships	Amadey and other malware relationships in OpenCTI	CornFlake and ChocoShell
Infrastructure context	AS202412 / OMEGATECH LTD relationship in selected campaign	Attacker-controlled infrastructure reached through captive-portal manipulation
Key defensive focus	Endpoint controls, user-execution monitoring, persistence and malware detection	Identity protection, phishing resistance, network visibility and authentication monitoring

Defensive considerations

- Monitor suspicious user-execution events, especially command execution initiated after browser activity.
- Use endpoint telemetry to detect unusual process creation, persistence and remote-access activity associated with malware execution.
- Monitor domains, URLs, IP addresses and file hashes from the investigation as intelligence inputs, while recognising that infrastructure may change.
- Strengthen identity protections around Microsoft 365 and Entra ID, with particular attention to phishing-resistant authentication and abnormal authentication flows.
- Monitor suspicious captive-portal, DNS and HTTP behaviour on corporate or travel-connected networks where network manipulation could redirect users.
- Use the ATT&CK; relationships as a basis for detection engineering, but validate each detection against the evidence and telemetry available in the target environment.
- Maintain connector and ingestion controls in OpenCTI so intelligence volume does not overwhelm the collection environment.

Detection-engineering note

The indicators captured in OpenCTI can support IOC-based detections, but the investigation also shows why behaviour-based detections matter. Disposable domains and changing infrastructure can age quickly, while user execution, process creation, authentication anomalies and persistence behaviours may remain useful detection signals.

10. Conclusion

The work started with the United States threat view in OpenCTI. Amadey (S1025) and UNC2452 were then taken forward for detailed analysis because they appeared first and second in that view.

The Amadey review linked the malware entity to multiple reports, indicators, malware relationships and ATT&CK; attack patterns. The selected Disposable Domains, Durable Hosting campaign showed fake CAPTCHA and ClickFix activity, disposable domains, infrastructure associated with AS202412, malware delivery, remote-access activity and persistence.

For UNC2452, the main focus was CaptiveCrunch. The evidence describes DNS and HTTP manipulation on captive portal networks, redirection to attacker-controlled infrastructure, Microsoft 365 credential theft, device-code phishing, ClickFix-based malware delivery, and relationships with CornFlake and ChocoShell.

For Fadurel Technologies, the two cases point to exposure around employees, endpoints, identities, cloud resources, online services and customer information. The findings can also be used as a starting point for detection engineering, threat hunting and further security assessment.

Overall assessment

The work provides a documented CTI view of two threats represented in the OpenCTI United States dataset and shows how the observed behaviour could be relevant to Fadurel Technologies. These findings should be revisited as the OpenCTI dataset and threat infrastructure change.



Appendix A. Evidence Register

ID	Evidence	Use in report
E01	vm-configuration.png	Final Ubuntu VM configuration
E02	open-cti-deployment.png	Docker/OpenCTI service health
E03	alienvault-otx-integration.png	OTX connector and ingestion queue
E04	united-states-location.png	Country selection: United States
E05	top-10-threats.png	United States threat landscape
E06	amadey-indicator-overview.png	Amadey OpenCTI overview
E07	amadey-related-reports.png	Amadey related reports
E08	amadey-indicator-list.png	Amadey indicators
E09	amadey-indicator-details.png	Amadey indicator details
E10	disposable-domains-durable-hosting.png	Amadey campaign report
E11	amadey-malware-indicator.png	Amadey malware relationships
E12	Attack-patterns-kill-chain.png	Amadey ATT&CK; relationship map
E13	captivecrunch-report.png	UNC2452 CaptiveCrunch report
E14	unc2452-malware.png	UNC2452 malware relationships
E15	unc2452-indicators.png	UNC2452 indicators
E16	unc2452-mitre-attack.png	UNC2452 ATT&CK; relationship map

Evidence files are retained in the project repository under the corresponding OpenCTI deployment, country threat landscape and threat-analysis screenshot directories.



Appendix B. ATT&CK Relationship Tables

This appendix preserves the broader ATT&CK; relationship maps recorded in the project. These are knowledge-graph relationships and are not equivalent to campaign-specific proof.

B.1 Amadey relationship map

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Resource Development	T1583	Acquire Infrastructure	Obtaining infrastructure that can support malicious operations.
Resource Development	T1583.004	Acquire Infrastructure: Server	Obtaining or renting servers for activities such as hosting, staging or C2.
Resource Development	T1588	Obtain Capabilities	Obtaining capabilities needed for an operation rather than developing everything internally.
Resource Development	T1588.001	Obtain Capabilities: Malware	Obtaining malware as a capability that can be used during an operation.
Resource Development	T1608	Stage Capabilities	Preparing or placing capabilities on infrastructure so they can later be used against targets.
Initial Access	T1185	Browser Session Hijacking	Abusing a browser session to intercept or inherit authenticated browser information such as cookies or sessions.
Initial Access	T1506	Web Session Cookie	Using stolen web-session cookies to authenticate to web applications without repeating the normal authentication process.
Execution	T1053	Scheduled Task/Job	Using scheduled tasks or jobs to execute malicious code at a particular time or repeatedly.
Execution	T1059	Command and Scripting Interpreter	Using command shells or scripting interpreters to execute commands or scripts.
Execution	T1106	Native API	Using operating-system APIs to perform actions such as creating processes or gathering information.
Execution	T1129	Shared Modules	Executing code by loading shared modules such as DLLs or shared objects.
Execution	T1204	User Execution	Relying on the victim to perform an action that causes malicious activity to execute.
Persistence	T1053	Scheduled Task/Job	Using scheduled execution to maintain recurring or persistent execution.
Persistence	T1112	Modify Registry	Modifying Windows Registry settings to support persistence, execution, configuration changes or defense evasion.
Persistence	T1136	Create Account	Creating an account that can provide continued access to a system or environment.
Persistence	T1543	Create or Modify System Process	Creating or modifying services, daemons or other system processes for persistent execution.
Persistence	T1547	Boot or Logon Autostart Execution	Configuring a system so malicious code automatically executes during boot or user logon.
Privilege Escalation	T1037	Boot or Logon Initialization Scripts	Using scripts that execute during system boot or logon, potentially providing persistence or elevated execution.
Privilege Escalation	T1053	Scheduled Task/Job	Abusing scheduled execution to run code under a more privileged context.
Privilege Escalation	T1055	Process Injection	Injecting malicious code into another process, potentially hiding execution or obtaining access to another process context.
Privilege Escalation	T1134	Access Token Manipulation	Manipulating access tokens to operate under another security context or obtain additional privileges.
Privilege Escalation	T1543	Create or Modify System Process	Creating or modifying system services/processes that may execute with elevated privileges.
Privilege Escalation	T1547	Boot or Logon Autostart Execution	Abusing autostart mechanisms that may execute with higher privileges.
Stealth	T1027	Obfuscated Files or Information	Obfuscating code, strings, files or other information to make analysis and detection more difficult.
Stealth	T1036	Masquerading	Making malicious files, processes or other objects appear legitimate.
Stealth	T1055	Process Injection	Hiding malicious execution inside another process.
Stealth	T1134	Access Token Manipulation	Manipulating security tokens to change the security context under which activity runs.

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Stealth	T1140	Deobfuscate/Decode Files or Information	Decoding or decrypting information that was previously obfuscated.
Stealth	T1218	System Binary Proxy Execution	Abusing trusted system binaries to execute malicious code.
Stealth	T1480	Execution Guardrails	Using environmental conditions to determine whether malicious activity should execute.
Stealth	T1497	Virtualization/Sandbox Evasion	Detecting or avoiding virtualized, sandboxed or analysis environments.
Defense Impairment	T1112	Modify Registry	Modifying Registry settings can be used to alter security-related configuration or otherwise impair defenses.
Credential Access	T1003	OS Credential Dumping	Obtaining credentials from operating-system locations such as credential databases or memory.
Credential Access	T1056	Input Capture	Capturing information entered by users, including potentially credentials.
Credential Access	T1528	Steal Application Access Token	Stealing application access tokens that can be used to access resources as the legitimate user or application.
Credential Access	T1539	Steal Web Session Cookie	Stealing authenticated web-session cookies so they can be reused to access applications.
Credential Access	T1552	Unsecured Credentials	Searching for credentials stored in insecure locations.
Credential Access	T1555	Credentials from Password Stores	Obtaining credentials from password stores such as browser or operating-system credential repositories.
Credential Access	T1568	Dynamic Resolution	Using dynamic mechanisms to locate or resolve attacker infrastructure.
Discovery	T1007	System Service Discovery	Enumerating services running or registered on a system.
Discovery	T1010	Application Window Discovery	Identifying open application windows to understand what applications are being used or identify security tools.
Discovery	T1012	Query Registry	Querying the Windows Registry to gather information about configuration, software or the operating system.
Discovery	T1018	Remote System Discovery	Identifying other systems on the network that may be useful for further activity.
Discovery	T1023	Shortcut Modification	Modifying shortcuts so that a malicious program or command executes when the shortcut is used.
Discovery	T1049	System Network Connections Discovery	Enumerating active network connections involving the compromised system.
Discovery	T1057	Process Discovery	Enumerating running processes on a system.
Discovery	T1082	System Information Discovery	Gathering information about the operating system, computer name, hardware or system configuration.
Discovery	T1083	File and Directory Discovery	Searching files, directories and filesystem locations for useful information.
Discovery	T1497	Virtualization/Sandbox Evasion	Checking for virtualization or sandbox indicators to determine whether the environment is suitable for execution.
Discovery	T1518	Software Discovery	Identifying installed software and software versions.
Lateral Movement	T1021	Remote Services	Using remote services or protocols to access other systems.
Collection	T1005	Data from Local System	Collecting information from the compromised local system.
Collection	T1056	Input Capture	Capturing information entered by the user.
Collection	T1113	Screen Capture	Taking screenshots of the victim's screen.
Collection	T1115	Clipboard Data	Collecting information stored in the clipboard.
Collection	T1119	Automated Collection	Automatically gathering information according to predefined collection logic.
Command and Control	T1071	Application Layer Protocol	Using application-layer protocols for command and control communication.
Command and Control	T1095	Non-Application Layer Protocol	Using protocols outside normal application-layer protocols for C2 communication.
Command and Control	T1105	Ingress Tool Transfer	Downloading files, tools or additional malware onto a compromised system.
Command and Control	T1210	Exploitation of Remote Services	Exploiting remote services to gain access to additional systems.
Command and Control	T1568	Dynamic Resolution	Dynamically resolving or locating C2 infrastructure.

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Command and Control	T1571	Non-Standard Port	Using unusual ports for network communication to make traffic harder to identify or filter.
Command and Control	T1573	Encrypted Channel	Encrypting C2 communications to prevent defenders from easily inspecting the traffic.
Exfiltration	T1041	Exfiltration Over C2 Channel	Sending stolen information through an existing command-and-control channel.



B.2 UNC2452 relationship map

Tactic shown in OpenCTI	ATT&CK; ID	Attack Pattern	What the relationship means
Stealth	T1027	Obfuscated Files or Information	Obfuscating files, code or other information to make analysis and detection more difficult.
Privilege Escalation	T1055	Process Injection	Injecting code into another process to execute malicious code and potentially evade defenses or elevate privileges.
Collection	T1056	Input Capture	Capturing user input that may contain credentials or other information.
Collection	T1056.001	Input Capture: Keylogging	Capturing keystrokes from a victim system.
Execution	T1059	Command and Scripting Interpreter	Using command or scripting interpreters to execute commands, scripts or binaries.
Execution	T1059.001	PowerShell	Using PowerShell to execute commands or scripts.
Execution	T1059.003	Windows Command Shell	Using the Windows Command Shell to execute commands or scripts.
Collection	T1113	Screen Capture	Capturing screenshots of a victim's screen.
Stealth	T1140	Deobfuscate/Decode Files or Information	Decoding or decrypting information that was previously obfuscated or encrypted.
Execution	T1204	User Execution	Relying on the victim to perform an action that causes malicious activity to execute.
Execution	T1204.003	User Execution: Malicious File	Relying on the victim to open or execute a malicious file.
Credential Access	T1528	Steal Application Access Token	Stealing application access tokens that can be used to access resources as a legitimate user or application.
Credential Access	T1539	Steal Web Session Cookie	Stealing authenticated web-session cookies for reuse in accessing web applications.
Privilege Escalation	T1548	Abuse Elevation Control Mechanism	Circumventing mechanisms designed to control privilege elevation.
Privilege Escalation	T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control	Bypassing Windows User Account Control to execute with elevated privileges.
Collection	T1557	Adversary-in-the-Middle	Positioning between networked devices to support activities such as credential theft, network sniffing or traffic manipulation.
Collection	T1557.002	Adversary-in-the-Middle: ARP Cache Poisoning	Manipulating ARP cache information to position the adversary between networked systems.
—	T1562	Impair Defenses	Attempting to weaken or interfere with security controls and defensive mechanisms.
—	T1562.001	Impair Defenses: Disable or Modify Tools	Disabling or modifying security tools or their configurations to reduce defensive visibility or protection.
Initial Access	T1566	Phishing	Using phishing techniques to gain access to victim systems.
Initial Access	T1566.002	Phishing: Spearphishing Link	Using malicious links to direct victims toward attacker-controlled content or resources.

B.3 Source and evidence notes

Primary project sources: OpenCTI knowledge objects and screenshots captured during the investigation; AlienVault OTX connector data; MITRE ATT&CK; relationships represented in OpenCTI; and the campaign reports selected within the investigation.

The report intentionally retains uncertainty where the project evidence is incomplete or truncated. No missing indicator values, labels or campaign techniques were expanded by inference.